4 Potential Explanations for Anomalies

- Numerous studies have tried to explain *why* the documented anomalies *exist* or why they *persist*.
  1. **Risk-based explanations** – argue that the 'seemingly anomalous' returns are not *abnormal*, but they compensate for risk.
  2. **Estimation error** – argue that the 'normal' return was not correctly measured, creating the 'appearance of an abnormal return'.
  3. **Market frictions** – focus primarily on why abnormal returns are not 'arbitraged away'. Do not necessarily explain how the abnormal return appeared in the first place.
  4. **Behavioral biases** – explore biases in investor behavior that may give rise to deviations from exact asset pricing.
- Explanations that focus on why the anomalies *exist* give either a *rational* or *irrational* reason why the seemingly abnormal return is there.
- Explanations that focus on why the anomalies *persist* do not necessarily rationalize why an anomaly is there in the first place, but *why it does not disappear*, i.e. why rational investors do not arbitrage away the anomous return.

Risk-Based Explanations

- Risk-based explanations argue that
  - the **positive returns** of the 'anomaly-based' strategy are **not abnormal**, but they **compensate for risk**.
  - the ***long side* is *riskier*** than the short side of the strategy
    $\Longrightarrow$ the ***long side should generate higher returns*** to compensate for the higher risk—i.e. the higher returns of the long side are not abnormal.
    $\Longrightarrow$ Once we properly adjust the returns of both sides of the strategy for risk, there are no abnormal returns.
- Risk-based explanations give a ***rational* reason for the *existence*** of the 'seemingly abnormal' returns.
- Some papers argue that risk-based explanations explain some, but not all, of the anomaly profits—the anomaly is ***partially explained by risk***.

- **Punchline**: **"alpha is beta is disguise"**, because:
  - there are **additional betas** to **new (previously omitted) factors**
  - **betas are different and/or time-varying**: they change with **market conditions**, or **firm-level characteristics**.

4.1 Risk-Based Stories: "Alpha is beta in disguise"

- In essence, **the characteristic the anomaly-based strategy was sorting on was 'proxying' for risk exposure**
  —what we thought was an **'alpha'** was basically a **'beta in disguise'**.

$$\begin{aligned} \alpha &= \overline{r_{10,t} - r_{1,t}} \\ &= \overline{\beta_{10,t} F_t - \beta_{1,t} F_t} \\ &= \overline{(\beta_{10,t} - \beta_{1,t}) F_t} \end{aligned} \tag{100}$$

  i.e. $\hat{\alpha} > 0$ because the true $\beta_{10} > \beta_{1,t}$.

- Some papers advocate that the market beta is different.
- Others argue that the alpha is driven not by the market beta, but by a beta with respect to a different factor.
- Or it could be that the risk exposure only **partially** explains the anomaly:

$$\begin{aligned} \alpha &= \overline{r_{10,t} - r_{1,t}} \\ &= \overline{[\alpha^*_{10} + \beta_{10,t} F_t]} - \overline{[\alpha^*_1 + \beta_{1,t} F_t]} \\ &= \alpha^* + \overline{(\beta_{10,t} - \beta_{1,t}) F_t} \end{aligned} \tag{101}$$

  i.e. the anomaly overstates the true abnormal returns: $\alpha > \alpha^*$.
  Specifically, part of the observed alpha is a beta in disguise.

4.2 Examples of Risk-Based Stories:

A. There are new (previously omitted) risk factors

1. The Fama-French SMB and HML Factors

- Fama and French (1993) argue that small stocks earn higher returns than big stocks, not because they are small, but because they have higher exposure to the SMB factor:

$$\beta_{Small,SMB} > \beta_{Big,SMB}$$

  Value stocks earn higher returns than growth stocks, not because they have higher BM ratios, but because they have higher exposure to the HML factor:

$$\beta_{Value,HML} > \beta_{Growth,HML}$$

- The characteristic (size, BM) does not impact returns directly, but it impacts betas, which in turn command a risk-premium.
- The profit is not 'anomalous' but 'compensation for risk.'

2. Carhart Momentum Factor – a.k.a UMD or MOM

- Carhart (1997) argues that the return differential between winners (P10) and losers (P1) is due to systematic risk. The factor returns are: $r_{UMD} = r_{P10} - r_{P1}$.

3. World Credit Risk Factor - CREDIT

- Avramov, Chordia, et al. (2012) argues that the higher return of emerging market countries (high credit risk) is due to exposure to (higher betas wrt) a world credit risk factor. Controlling for exposure to CREDIT, emerging market and high credit risk countries do not earn abnormal returns.
- The CREDIT factor is constructed as the return differential between the worst and best rated countries: $r_{CREDIT} = r_{C3} - r_{C1}$.
- **Exhibit A compares results from portfolio sorts** using raw and risk-adjusted returns, where returns are risk-adjusted as in Brennan, Chordia, and Subrahmanyam (1998) by running time-series regressions of each individual country excess return on the respective factors. The risk-adjusted returns, r^*_{t+1}, are the intercept and residual from these time-series regressions.
- **Exhibit B shows cross-sectional regressions** using r and r^*.
- **Exhibit C shows Gibbons, Ross, and Shanken (1989) TSR tests.** The efficiency of the CREDIT factor cannot be rejected, while alternatives are.
- **Exhibit D shows Fama-MacBeth regressions.** CREDIT is priced.

• Exhibit A: Portfolio sorts: sorting on country's credit rating

	Sovereign Rating Group (C1=Lowest, C3=Highest Risk)			
	C1	C2	C3	C3-C1
Raw returns				
r_{t+1}	0.84	0.80	1.41	0.57
	(2.71)***	(2.36)**	(3.58)***	(2.15)**
Adjusting for world market [MKT] and foreign exchange risk factors				
r^*_{t+1}	0.16	0.13	0.75	0.58
	(1.54)	(0.67)	(2.62)***	(2.20)**
Adjusting for Fama and French (1998) international MKT and HML Factors				
r^*_{t+1}	0.23	0.14	0.82	0.58
	(1.78)*	(0.66)	(2.71)***	(2.16)**
Adjusting for the FF98 MKT, HML, and international momentum factors				
r^*_{t+1}	0.25	0.19	0.89	0.63
	(1.85)*	(0.90)	(2.81)***	(2.24)**
Adjusting for the MKT and Pástor and Stambaugh (2003) US liquidity factors				
r^*_{t+1}	0.54	0.49	1.10	0.56
	(4.69)***	(2.40)**	(3.81)***	(2.09)**
Adjusting for the CREDIT factor				
r^*_{t+1}	0.20	-0.09	0.30	0.10
	(1.79)*	(-0.50)	(2.27)**	(1.42)

• Exhibit B: Cross-sectional regressions: on lagged credit rating

Specification	*Constant*	$Rating_t$	$Rating_t \times EmDummy$
Raw Returns: r_{t+1}			
1	0.33	0.07	
	(1.01)	(2.06)**	
3	0.27	0.10	-0.04
	(0.85)	(1.82)*	(-0.76)
Risk-Adjusted Returns: $r^*_{t+1}\|[MKT, FOREX]$			
1	-0.02	0.07	
	(-0.16)	(2.14)**	
3	-0.04	0.08	-0.02
	(-0.29)	(1.79)*	(-0.37)
Risk-Adjusted Returns: $r^*_{t+1}\|[FF98 \quad MKT, HML]$			
1	0.03	0.07	
	(0.22)	(2.05)**	
3	-0.00	0.09	-0.03
	(-0.01)	(1.77)*	(-0.57)
Risk-Adjusted Returns: $r^*_{t+1}\|[MKT, FX, CREDIT]$			
1	-0.00	0.03	
	(-0.03)	(1.44)	
3	-0.05	0.06	-0.04
	(-0.38)	(1.12)	(-0.82)

- **Exhibit C: Gibbons, Ross, and Shanken (1989) time-series test of the efficiency of a factor**

Factors	J-statistic	p-value	95% Critical Value
MKT	1.92	0.00	1.50
FOREX	13.90	0.00	1.50
MKT+FOREX	14.11	0.00	1.51
FF93 MKT+HML	1.64	0.03	1.51
EMERG	1.57	0.03	1.50
CREDIT	**0.96**	**0.55**	**1.50**

The GRS sample test statistic is: $J = \frac{T-N-K}{N}\left[1+\hat{E}(f)'\hat{\Omega}^{-1}\hat{E}(f)\right]\hat{\alpha}'\hat{\Sigma}^{-1}\hat{\alpha}$, where $J \sim F_{N,T-N-K}$, $\hat{\alpha} = [\hat{\alpha}_1, \hat{\alpha}_2, ..., \hat{\alpha}_N]'$ are the estimated intercepts from individual time-series regressions and $\hat{\epsilon}_t = [\hat{\epsilon}_t^1, \hat{\epsilon}_t^2, ..., \hat{\epsilon}_t^N]'$ for $t = [1, 2, ..., T]$ are the estimated residuals, $\hat{\Sigma} = \frac{1}{T}\sum_{t=1}^{T}\epsilon_t\epsilon_t'$ is the residual covariance matrix, $\hat{\Omega}$ is the estimated covariance matrix of the factors, K is the number of factors, N is the number of test assets, and T is the number of time-series observations. The test assets are country equity returns with non-missing observations from January 1999 to December 2009: a total of 52 countries.

- **Exhibit D: The CREDIT factor is priced: its risk-premium averages 80 bps per month**

Asset-Pricing Tests using the Fama-MacBeth Method

1. Obtain betas from time-series regressions of country equity returns on the factors

$$R_{it} = a_i + \beta_{i,MKT} MKT + \beta_{i,FX} FX_t + \cdots + \beta_{i,CREDIT} CREDIT_t$$

2. Obtain risk premiums from cross-sectional regressions of country excess returns on the betas

$$R_{it} = a_t + \lambda_{t,MKT} \widehat{\beta}_{i,MKT} + \lambda_{t,FX} \widehat{\beta}_{i,FX} + \cdots + \lambda_{t,CREDIT} \widehat{\beta}_{i,CREDIT}$$

	Constant	λ_{MKT}	λ_{FX}	λ_{CREDIT}	λ_{EMERG}	$Adj.R^2$
1	0.13	0.72				5.51
	(0.37)	(1.56)				
2	0.22	0.73	-0.02			8.34
	(0.61)	(1.59)	(-0.27)			
5	0.28	0.42	0.03	0.80		14.35
	(0.78)	(0.94)	(0.42)	(2.33)**		
6	0.28	0.40	0.02	0.73	0.14	18.86
	(0.79)	(0.88)	(0.32)	(1.97)**	(0.35)	
7	0.52				0.82	6.43
	(1.93)*				(2.26)**	

'MKT' = world market factor 'FX' = foreign exchange factor 'EMERG' = Emerging Markets Factor

B. Risk changes with market conditions

$$Er_{i,t} = E(\beta_t r_{m,t}) = E(\beta_t)E(r_{m,t}) + cov(\beta_t, r_{m,t}) \tag{102}$$

- The *conditional* CAPM may be valid, but if we test it unconditionally, $cov(\beta_t, r_{m,t})$ may look like α_i.
- Petkova and Zhang (2005) [henceforth, PZ] evaluated the conditional CAPM conditioning both betas and the risk premium on macro-level information.
- **The conditioning variables** (standard in the time-series predictability literature) are **the dividend yield (DIV), the default spread (DEF), the term spread (TERM), and the Treasury bill rate (TB).**

- PZ estimate the risk premium in a time-series regression:

$$r_{mt+1} = \delta_0 + \delta_1\, DIV_t + \delta_2\, DEF_t + \delta_3\, TERM_t + \delta_4\, TB_t + e_{mt+1} \quad (103)$$

The conditional risk-premium is then:

$$\widehat{\gamma}_t = \widehat{\delta}_0 + \widehat{\delta}_1\, DIV_t + \widehat{\delta}_2\, DEF_t + \widehat{\delta}_3\, TERM_t + \widehat{\delta}_4\, TB_t \quad (104)$$

- PZ estimate conditional betas through conditional market regressions:

$$r_{it+1} = \alpha_i + (b_{i0} + b_{i1}\, DIV_t + b_{i2}\, DEF_t + b_{i3}\, TERM_t + b_{i4}\, TB_t) r_{mt+1} + \epsilon_{it+1} \quad (105)$$

The conditional betas are:

$$\widehat{\beta}_t = \widehat{b}_{i1} + \widehat{b}_{i1}\, DIV_t + \widehat{b}_{i2}\, DEF_t + \widehat{b}_{i3}\, TERM_t + \widehat{b}_{i4}\, TB_t \quad (106)$$

- They then examine how betas, $\widehat{\beta}$, vary with the risk premium, $\widehat{\gamma}$.
- PZ find that **"value betas tend to covary positively, and growth betas tend to covary negatively with the expected market risk premium."**
- **PZ conclude that 'value is riskier than growth'—providing a risk-based explanation for an asset pricing anomaly.**

- The value premium, α_{Value} is compensation for risk:

$$\begin{aligned}
\alpha_{Value} &= E\left[r_{Value,t} - r_{Growth,t}\right] \\
&= E[r_{Value,t}] - E[r_{Growth,t}] \\
&= E[\beta_{10,t} R_{m,t}] - E[\beta_{1,t} R_{m,t}] \\
&= E[\beta_{10,t}]\, E[R_{m,t}] + cov(\beta_{10,t}, R_{m,t}) - E[\beta_{1,t}]\, E[R_{m,t}] - cov(\beta_{1,t}, R_{m,t}) \\
&= (E[\beta_{10,t}] - E[\beta_{1,t}])\, E[r_{m,t}] + [cov(\beta_{10,t}, R_{m,t}) - cov(\beta_{1,t}, R_{m,t})] \quad (107)
\end{aligned}$$

- In other words, the returns of value stocks should be larger due to their higher covariance with the market.
- However, Lakonishok, Shleifer, and Vishny (1994) find that value stocks do much better than growth stocks in periods of poor market returns, arguing that value cannot be riskier than growth.
- PZ stress that ***realized* market returns are poor and noisy proxies for the *expected market premium***. In fact, when the *ex-ante* expected risk premium increases, the *ex-post* realized return is negative.

- Using the expected market risk premium, $\hat{\gamma}_{t+1}$, PZ find that value stocks do poorly in bad *expected* market return states, which makes them riskier.

$$r_{HML,t+1} = \alpha_{HML} + \beta_{HML,u}\, r_{m,t+1}\, d + \beta_{HML,d}\, r_{m,t+1}\, (1-d) + \epsilon_{HML,t+1}$$

  where $d = \mathbf{1}_{\hat{\gamma}>\bar{\gamma}}$.

- PZ find that HML beta is -0.26 in good times (when the market risk premium is above average, $\hat{\gamma} > \bar{\gamma}$) and 0.34 in bad times.

- PZ admit that their argument explains some but not all of the value premium.

C. Risk changes with firm characteristics

- Avramov and Chordia (2006) [henceforth, AC] examine pricing abilities of conditional and unconditional models, by **conditioning beta both on macro-level information** and **on firm-level characteristics**—beta is time-varying, changing with aggregate and firm-level conditions.
- **AC argue that firm-level characteristics impact betas. Characteristics impact returns through their impact on betas.**
- AC use their model to study whether size, book-to-market, momentum, and turnover can still predict cross-sectional differences in returns once the impact of these characteristics on betas is accounted for.
- AC allow beta to vary with size, BM, and past returns, as well as with macroeconomic variables:
  - First, for each security, they estimate a time-series specification of the form

$$r_{it} = \alpha_i + \underbrace{\beta_i(\mathbf{z}_{t-1}, \mathbf{z}_{it-1})'}_{conditioning} f_t + \epsilon_{it} \tag{108}$$

    where $\mathbf{z}_t$ **is a macro-level conditioning information** and $\mathbf{z}_{it}$ **denotes a firm-level characteristic**, e.g. size, book-to-

market.

- **The risk-adjusted return is obtained as the intercept and residual from (111)**

$$r^*_{i,t} = \alpha_i + \epsilon_{it} \tag{109}$$

- They then consider **cross-sectional regressions of risk-adjusted returns on firm-level size, book-to-market, past return variables, and turnover.**
- If the pricing model does a good job in pricing individual stocks then the slope coefficients in the cross-sectional regressions should be statistically indistinguishable from zero $\longmapsto$ the slope coefficients are, however, highly significant.
- That is, firm-level size, book-to-market, past return, and turnover are still important even when stock returns are adjusted by the SMB, HML, WML, and liquidity factors.

- AC find that **with constant betas, none of the models examined captures the three anomalies.**
- **When beta is allowed to vary, size and value effects are often explained, but the momentum anomaly remains robust.**

- **The momentum effect results from model mispricing that varies with macroeconomic variables.**